

Discovering Leitmotifs in Multidimensional Time Series

Research track: Time Series Data III

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Leitmotifs

- A **leitmotif** (leading motif) is a recurring theme or „motif“
- It carries symbolic significance in all forms of art, e.g. literature, movies, and music
- Humans associate leitmotifs with meaning, which enhances narrative cohesion and establishes emotional connections with the audience
 - **Richard Wagner** used these as a compositional technique in his operas

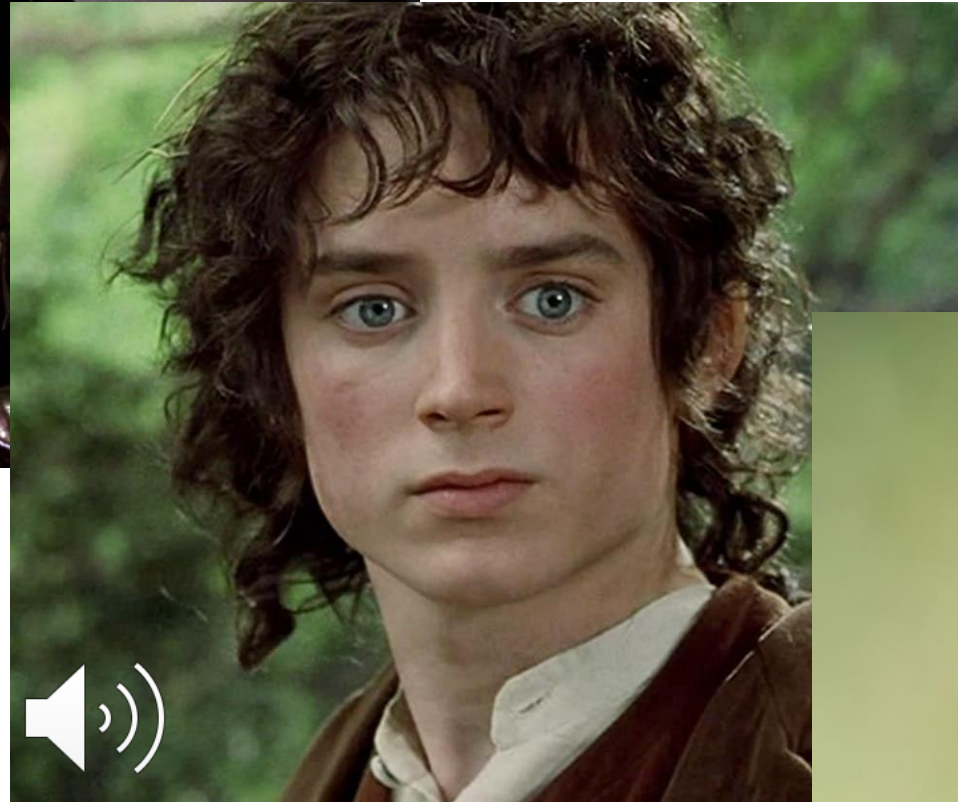


Ride of the Valkyries

John Williams – Imperial March



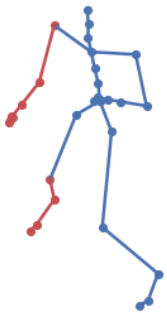
Howard Shore – The Shire



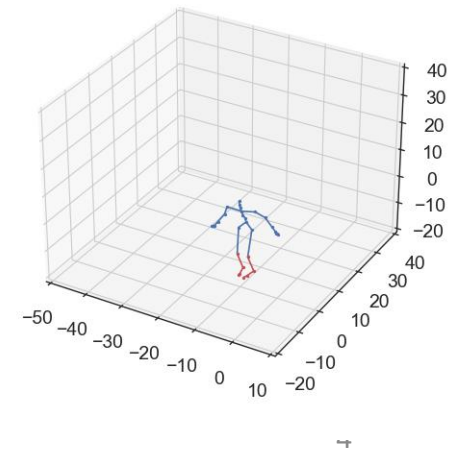
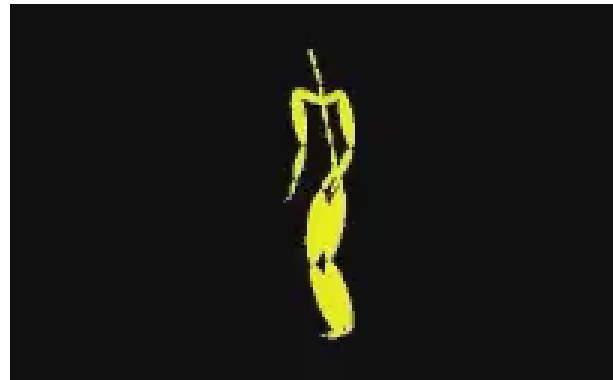
Common Starling



Motion Capture: Boxing Routine



Motion Capture: Charleston Dancing

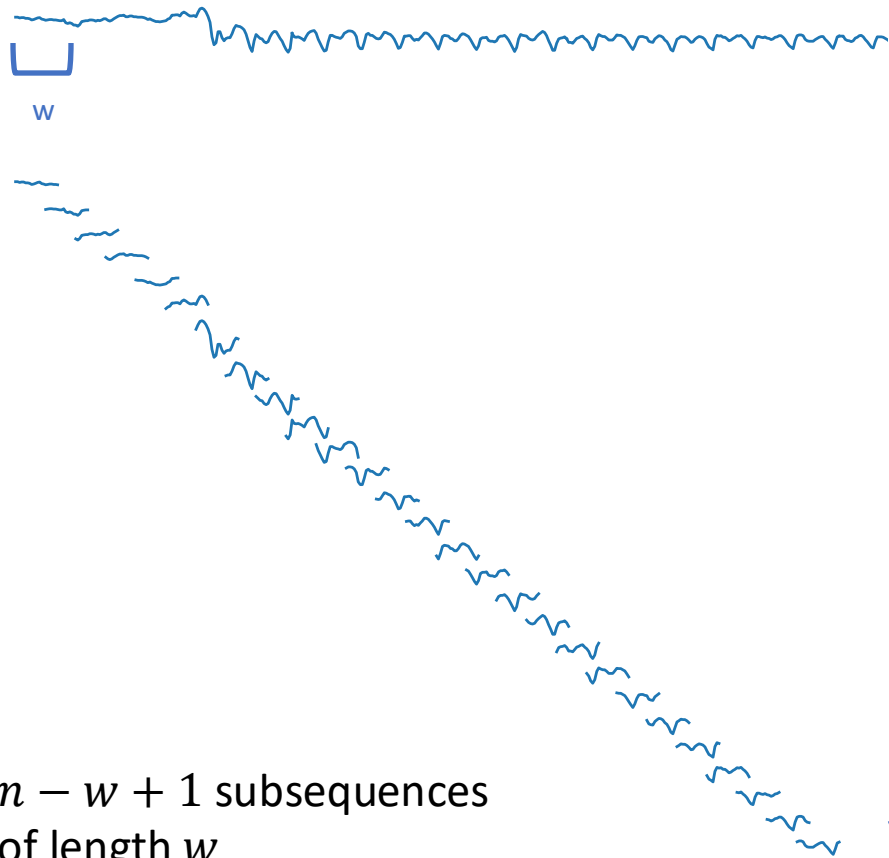


Challenging Problem

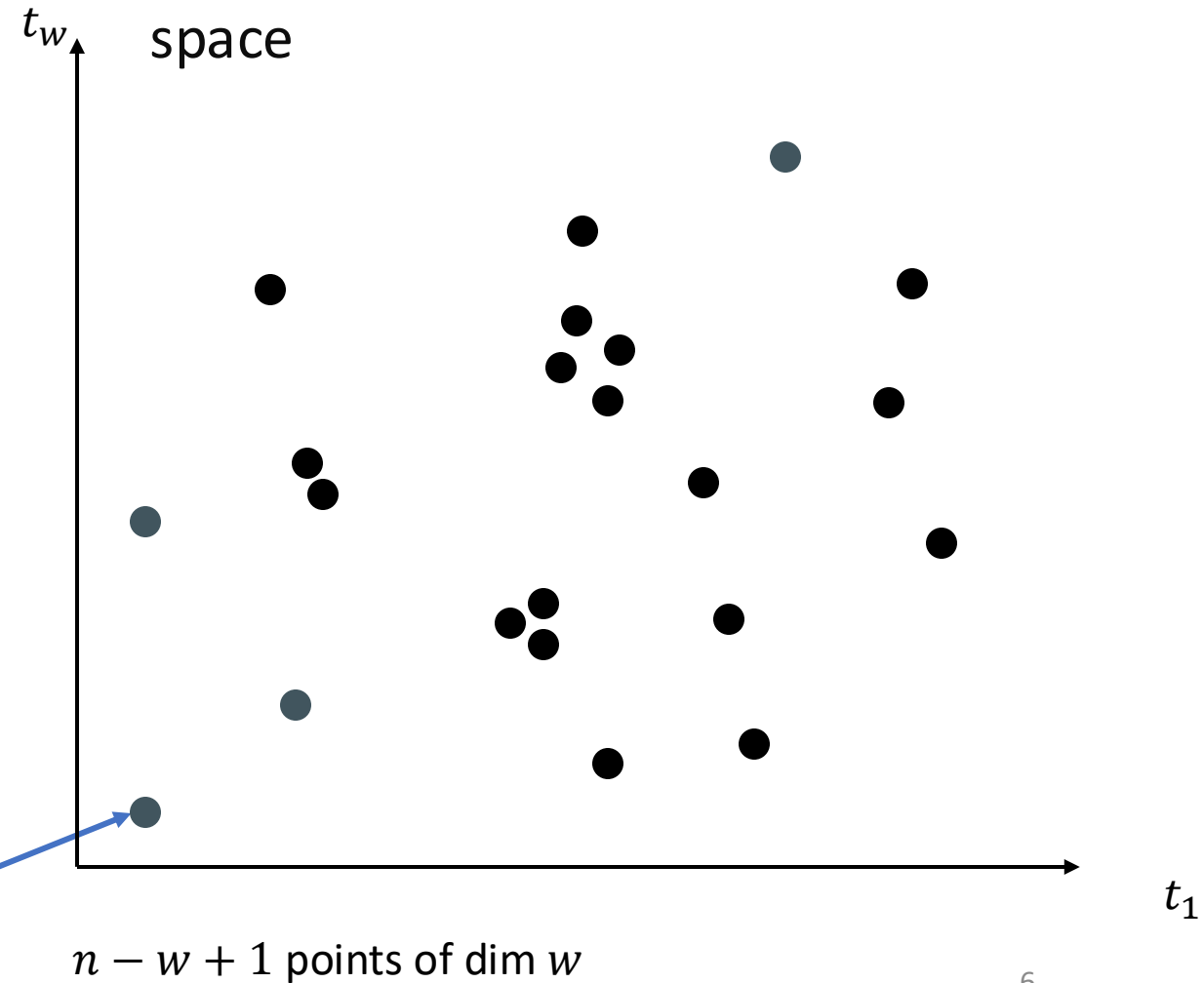
- A **leitmotif** is a **sub-dimensional motif set** of a time series with **symbolic significance** which
 - has unknown length l ,
 - occurs an unknown number of times k , and
 - occurs in an unknown number of sub-dimensions f
- It is an unsupervised and mathematically ill-defined problem of high complexity

Background: High-Dimensional Vector Space

We use a sliding window to extract w -length subsequences

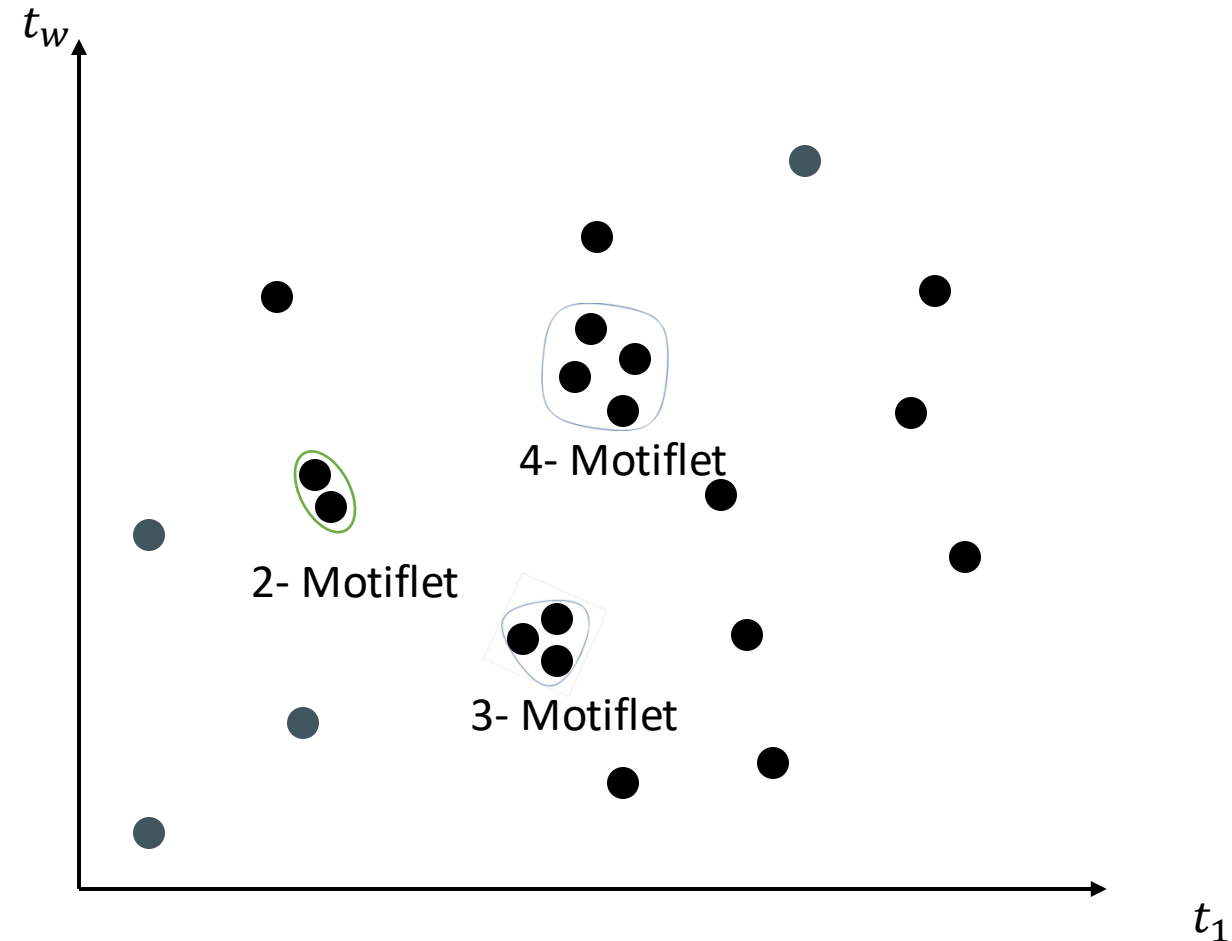


... and interpret these as points in w -dim space



Motiflets – VLDB`24

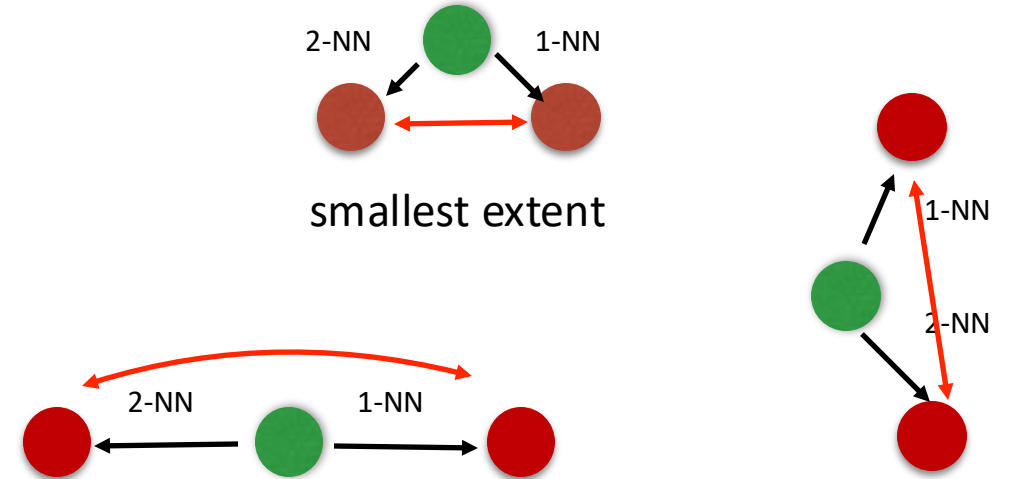
- Goal: Find smallest sets of similar points
- Motiflets: find k-frequent motif sets with **smallest pairwise distance (aka extent)**



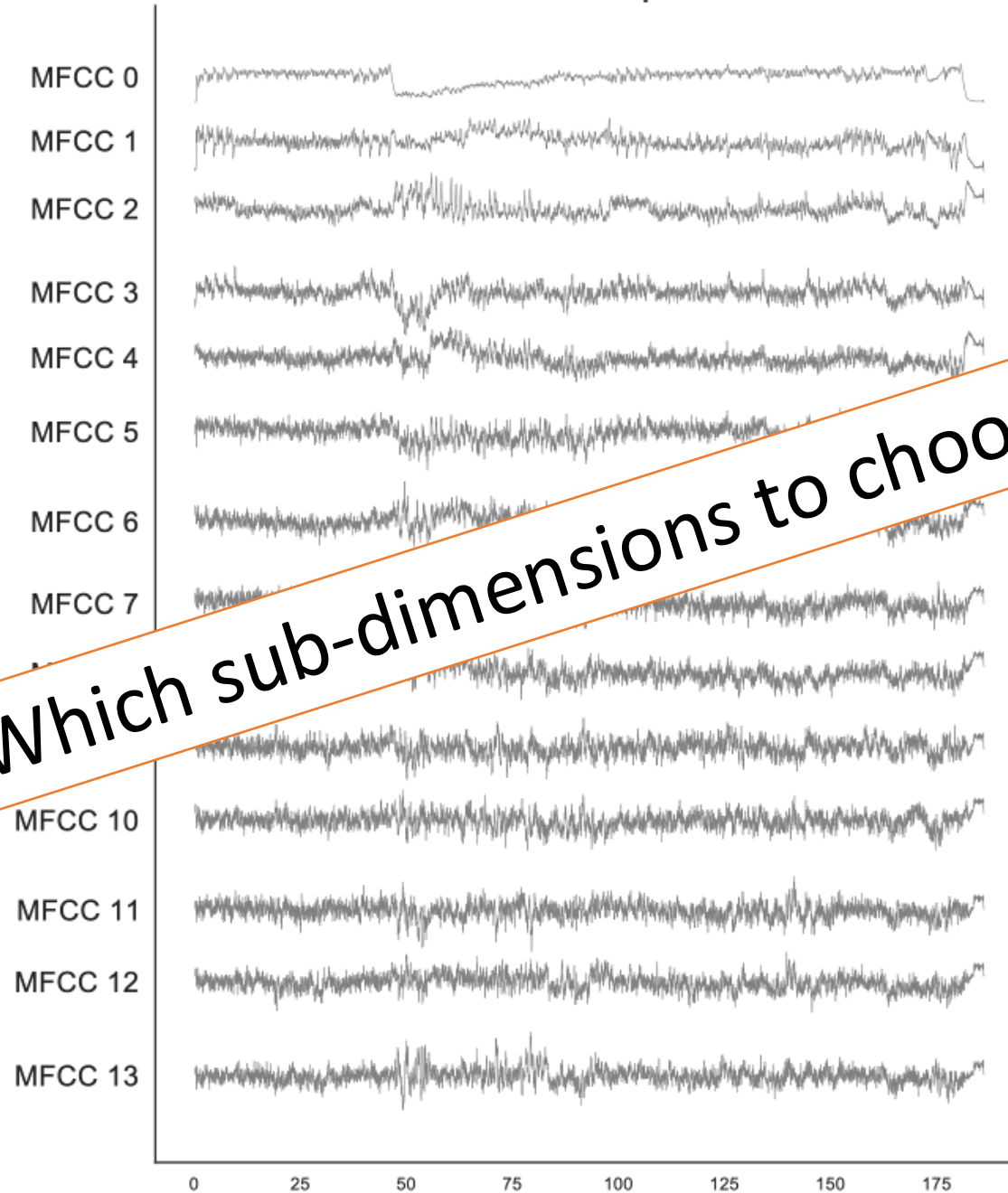
Motiflets – Algorithm

Input: freq. k and length l

1. Perform a k -NN search around each of $O(n)$ subsequences (*red dots*)
 2. For each candidate, compute maximum over pairwise distances (“*extent*”)
 3. Return: Minimum over all candidate sets
- Complexity: $O(k \cdot n^2)$
 - **But: Only applicable for one-dimensional data**



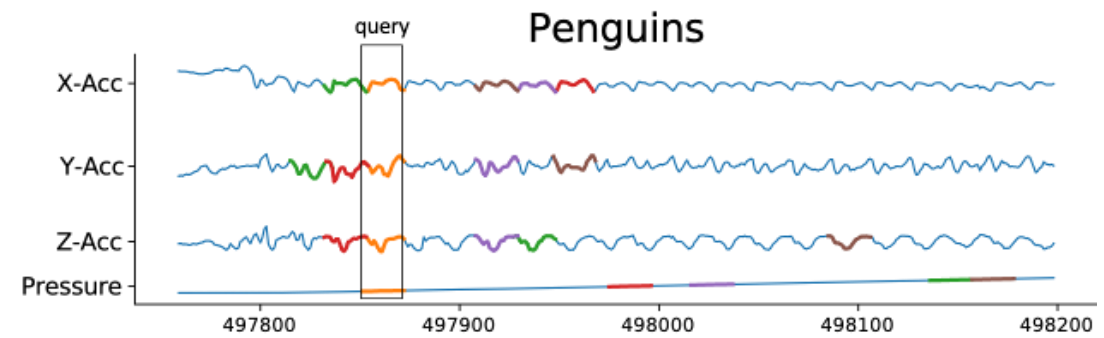
Star Wars - The Imperial March



Subdimensional Motif Sets

- Searching optimal Motif Sets along sub-dimensions is NP-hard
- We must enumerate all subsets of f out of d -dims
 - There are $\binom{d}{f}$ – many combinations
- We propose an approximate solution: LAMA

LAMA



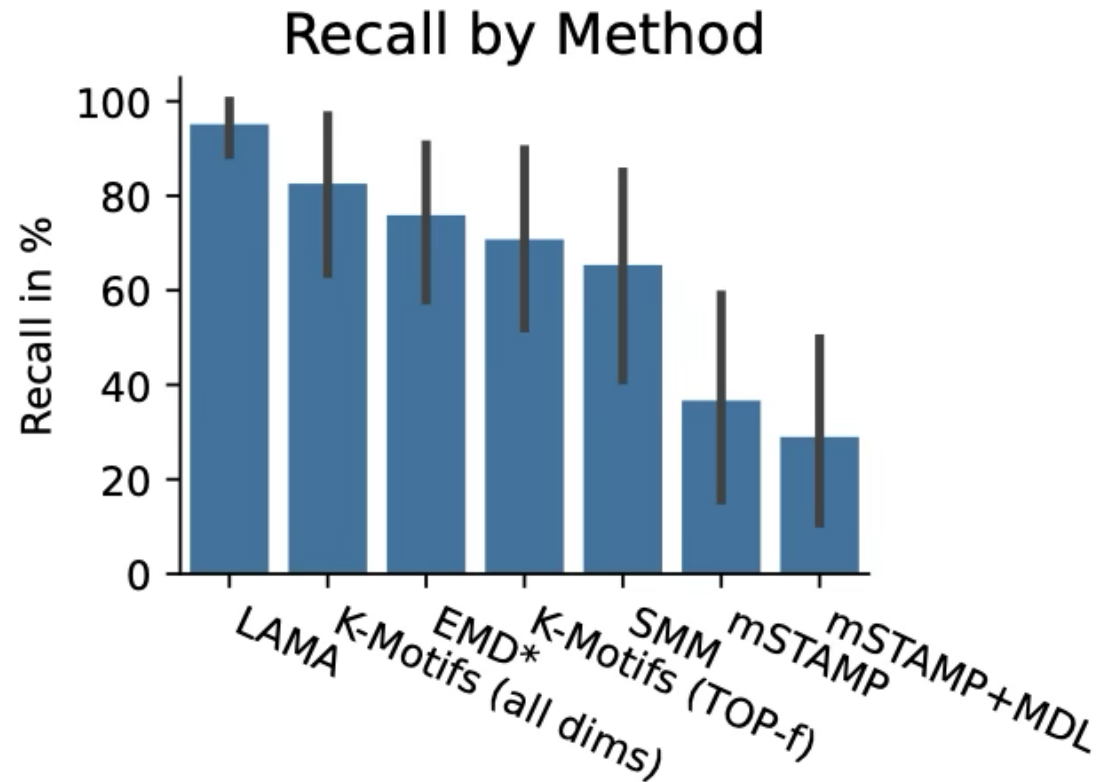
- For each a query subsequence:
 1. perform k-NN search along each dimension
 2. sort dimensions by k^{th} NN distance
 3. take f smallest values
 4. use *offsets* from dimension with smallest distance
 5. compute *extent* as sum over f dims
- Minimize extent over all f-dim candidates
- Worst / Best Case: $O(d k n^2) / O(d n^2)$

Experiments

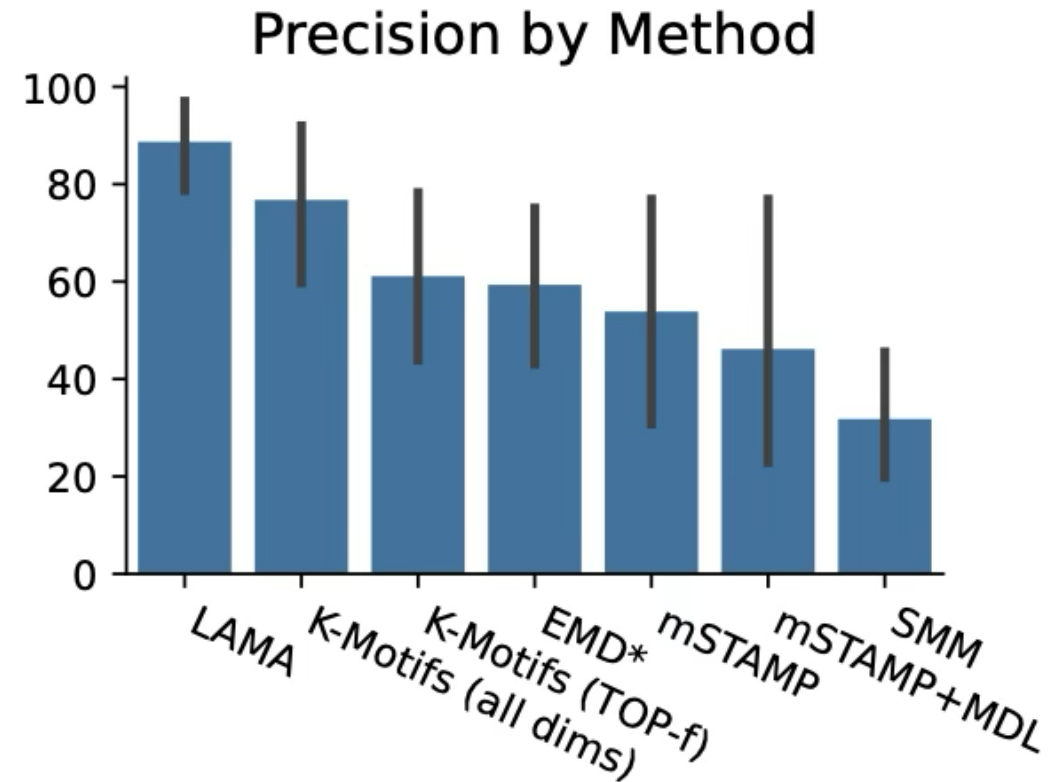
- We evaluated LAMA on 14 real-world use cases
- Four competitors
 - **SMM**: subdimension motif set discovery
 - **mSTAMP**: subdimensional pair motif discovery
 - **EMD**: PCA for channel reduction, followed by univariate motif set discovery
 - **K-Motifs**: aggregate distances for first f or all channels
- Each real-world use case has ground truth leitmotifs manually annotated
 - We can measure Precision/Recall

Use Case	Category	Length	Dim.	GT
Charleston [9]	Motion Capture	506	93	3
Boxing [9]	Motion Capture	4840	93	10
Swordplay [9]	Motion Capture	2251	93	6
Basketball [9]	Motion Capture	721	93	5
LOTR - The Shire	Soundtrack	6487	20	4
SW - The Imperial March	Soundtrack	8015	20	5
RS - Paint it black	Pop Music	9744	20	10
Linkin Park - Numb	Pop Music	8018	20	5
Linkin P. - What I've Done	Pop Music	8932	20	6
Queen - Under Pressure	Pop Music	9305	20	16
Vanilla Ice - Ice Ice Baby	Pop Music	11693	20	20
Starling [5]	Wildlife Rec.	2839	20	4
Physiodata [24]	Wearable Sensors	5526	5	20
Bitcoin Halving	Crypto/Stocks	3591	7	3

Lama outperforms competitors

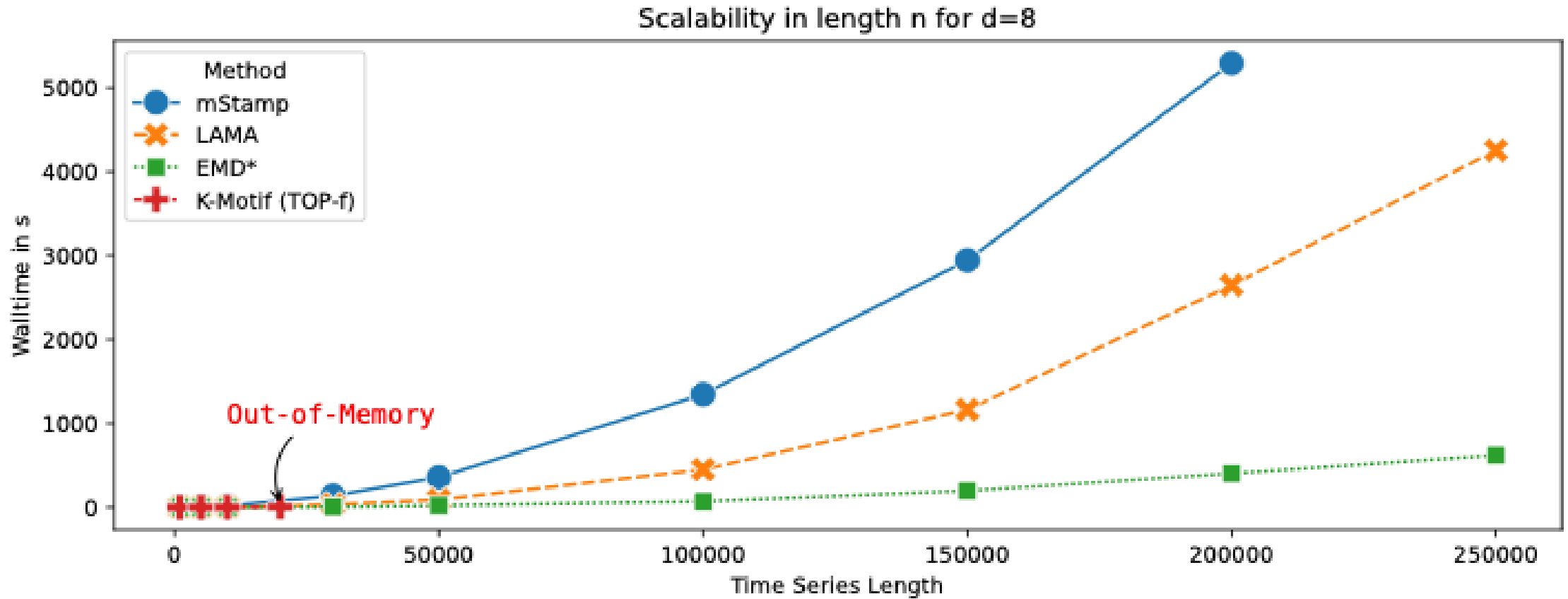


Most occurrences are found



If reported, it is correct

Scalability



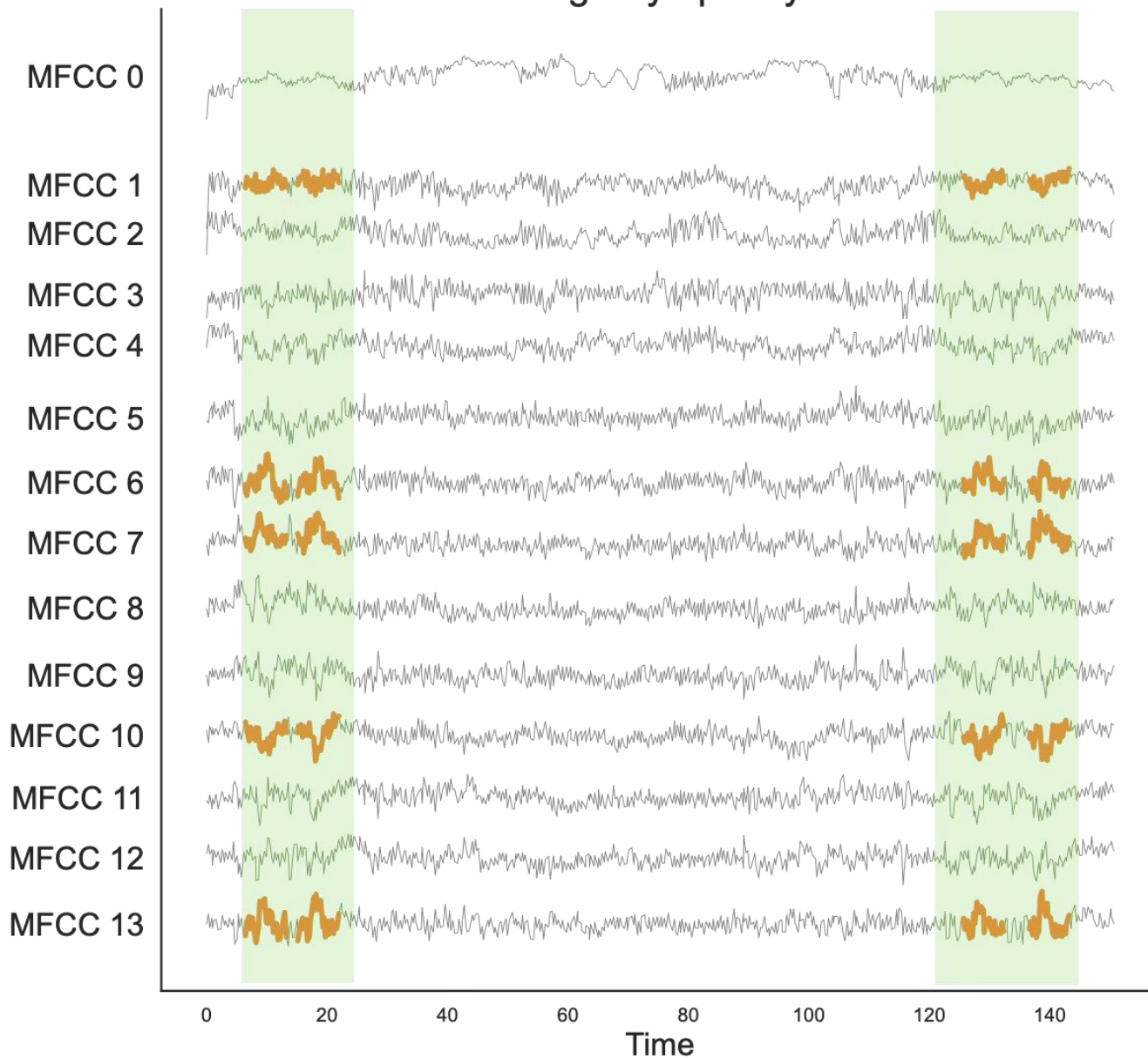
LAMA scales better than most methods but EMD

Use Case – LOTR

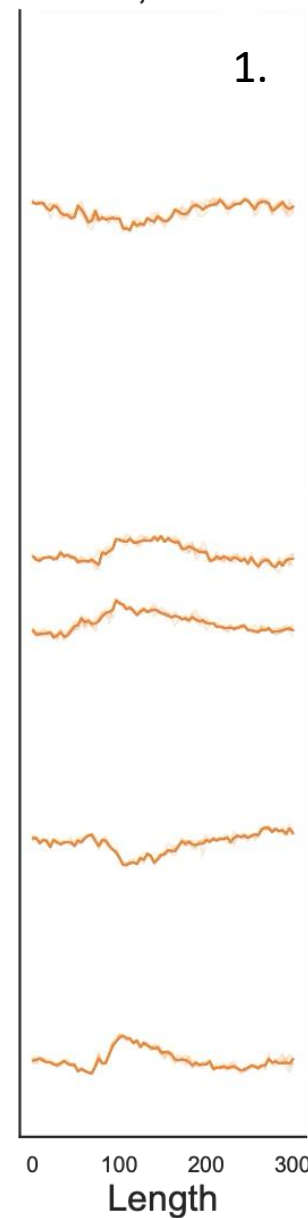
- The Shire was arranged by Howard Shore for Lord of the Rings
- It opens and ends with the Hobbits' leitmotif, played by a solo tin whistle
 - It manifests in a distinct pattern in several, but not all dimensions



Lord of the Rings Symphony - The Shire



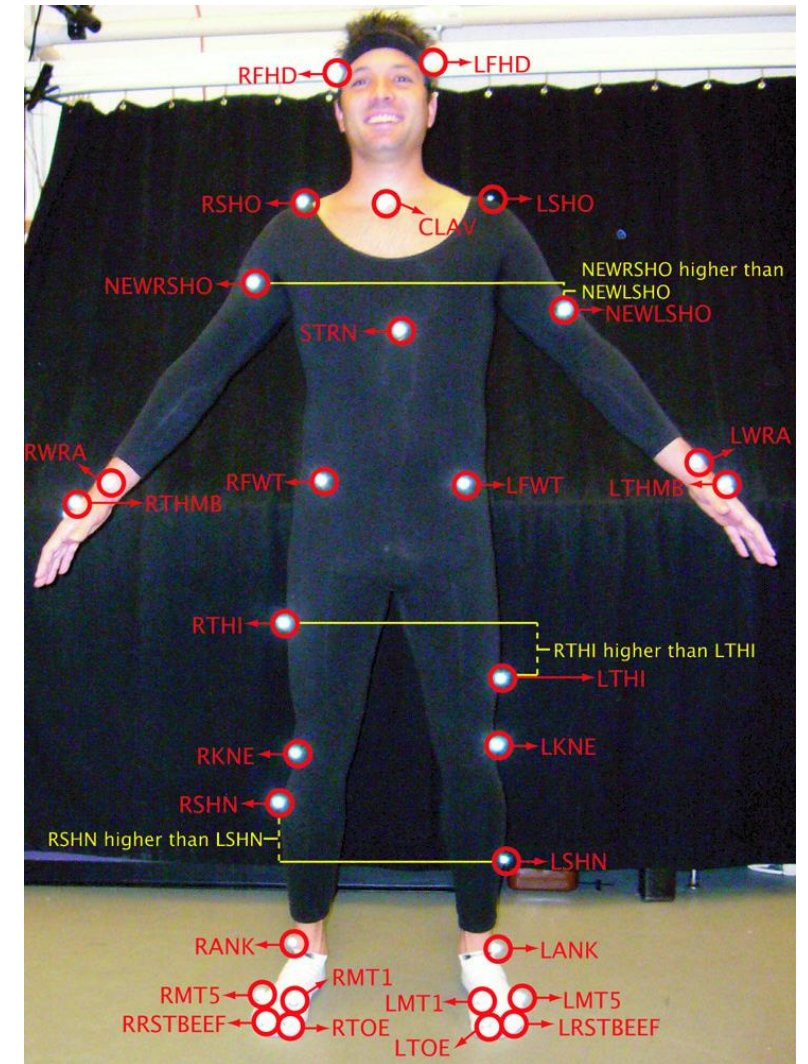
LAM
k=4, l=30



Tin Whistle playing at ~6 sec

Use Case – Motion Capture

- Data recorded by the the CMU Database
 - <http://mocap.cs.cmu.edu/>



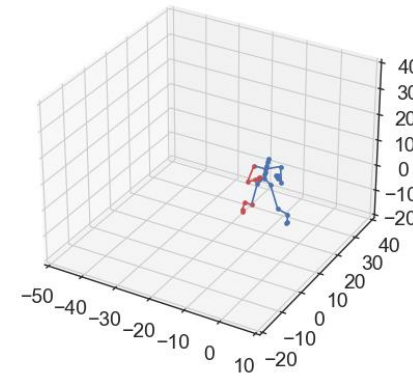
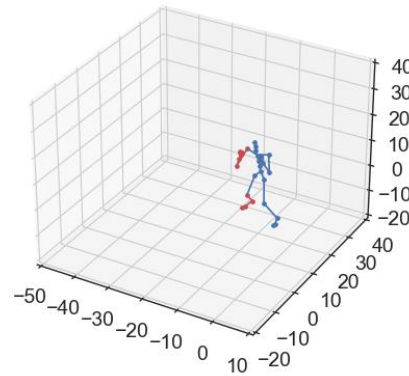
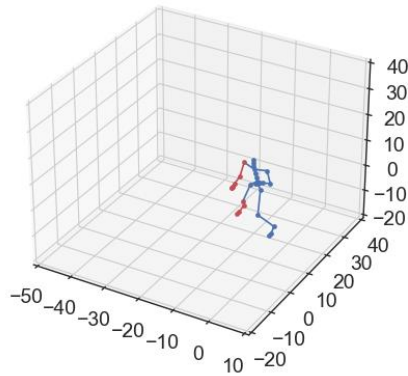
Use Case – Boxing

- We looked into boxing motion capture, of 41 sec at 120 frames per second
- We concentrate on right body joints:
 - 'rclavicle', 'rhumerus',
 - 'rradius', 'rwrist',
 - 'rhand', 'rfingers', 'rthumb',
 - 'rfemur', 'rtibia', 'rfoot', 'rtoes'



Use Case – Motion Capture

- We found 10 repeats of the leitmotif of length 0.8 sec
- It is the punching motion
- Here, showing only the first three



20's Charleston

- Originated in Charleston South Carolina
- In Europe, the dance became known mainly through the singer, dancer and actress Josephine Baker

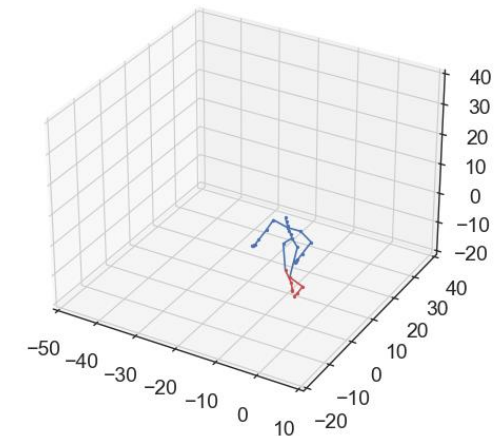
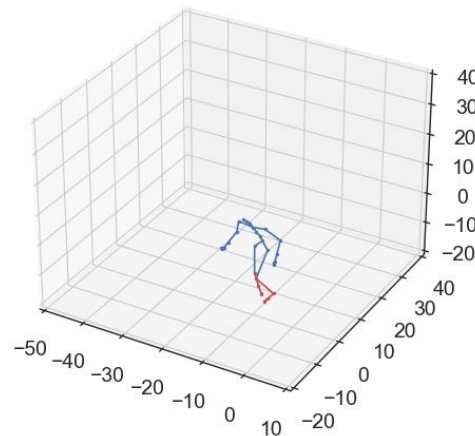
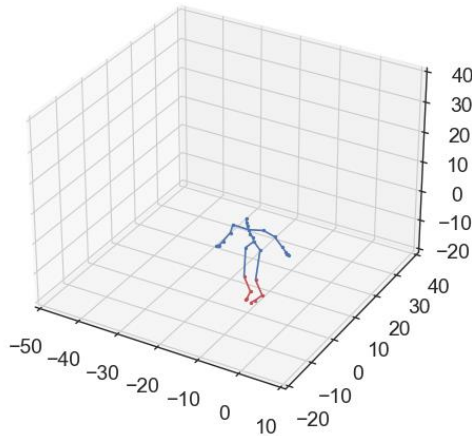


20's Charleston

- We use LAMA to find the basic step / footwork from motion data of a Charleston dancing routine
- The used routine contains 3 repeats of this leitmotif
- The found leitmotif is the basic step



3 repeats



Conclusion

- We present the first annotated Leitmotif benchmark
- Our experimental evaluation shows that LAMA has superior performance in detecting semantically meaningful patterns
- Our findings demonstrate that summarizing a time series through its leitmotifs offers a profound understanding of underlying patterns



**Source
Codes**

Thank you!
Questions?

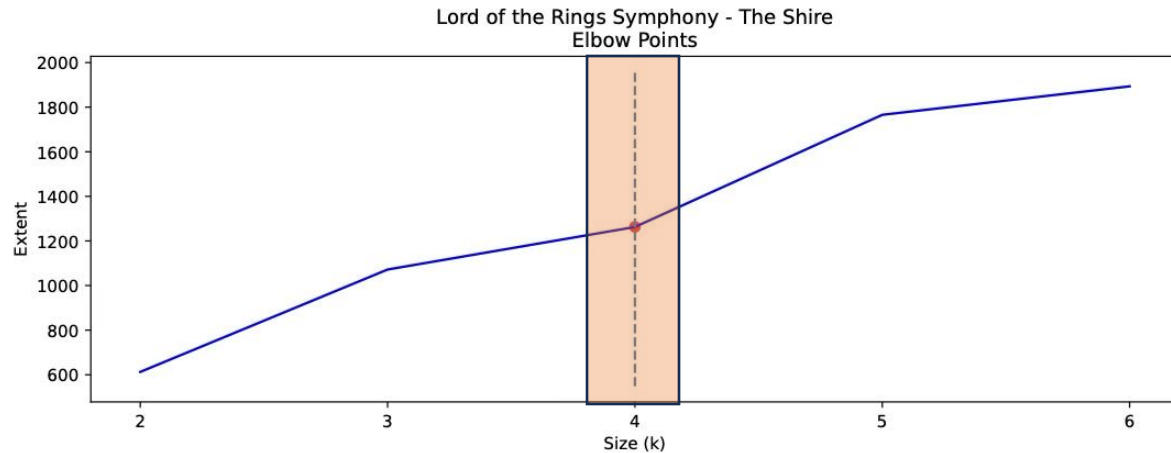


**Source
Codes**

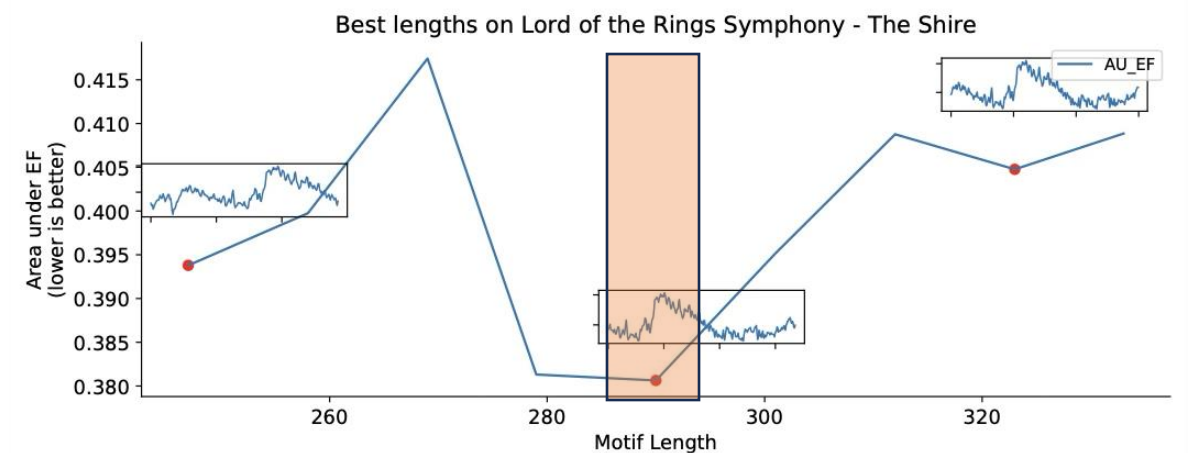
Backup Slides

Learning k and l

- Optimum at 4 repeats of roughly 7 seconds

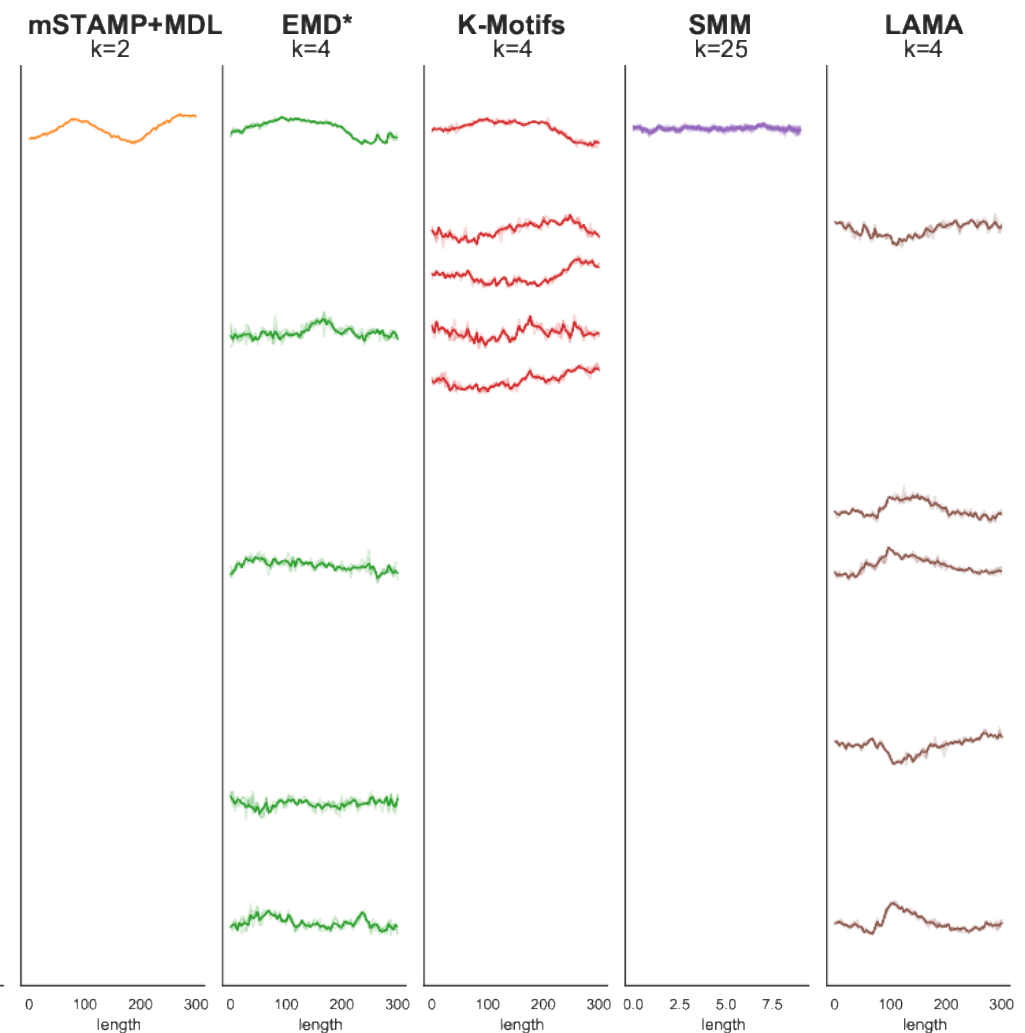
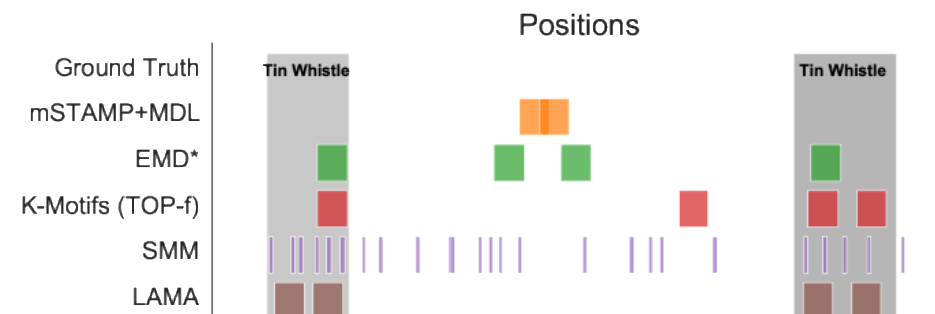
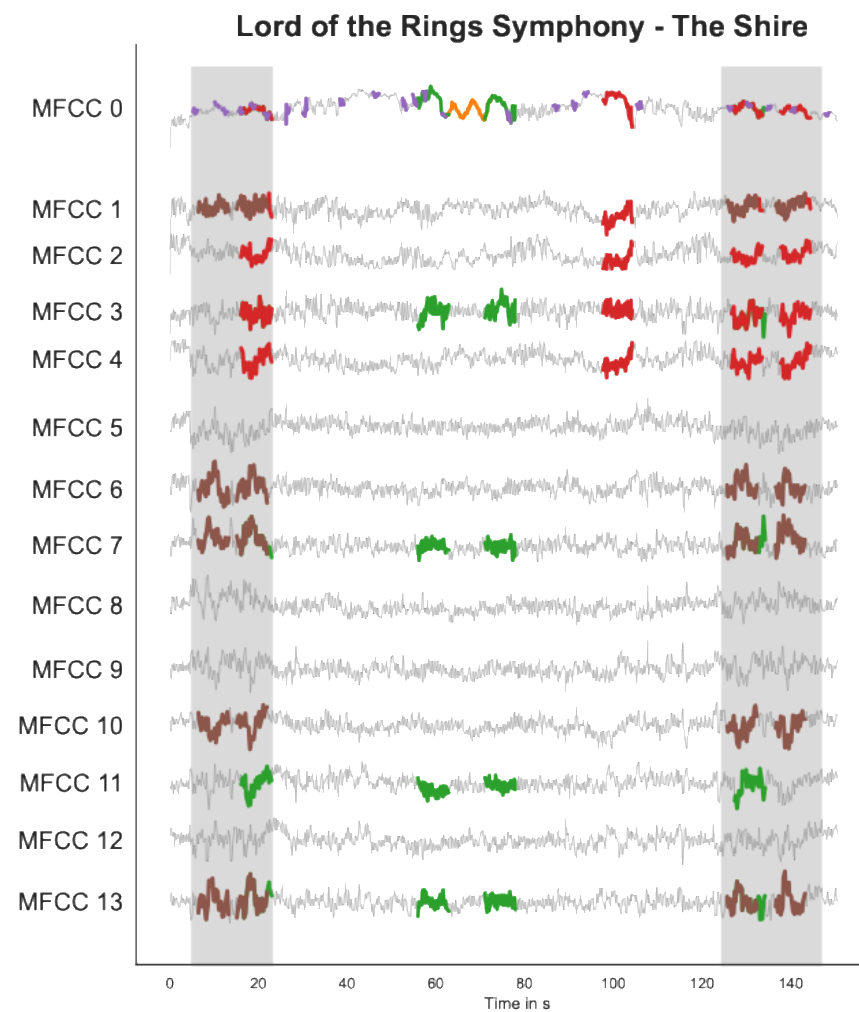


Elbow Points



Area under the Extent Function

- Details omitted for brevity – see paper



Leitmotif: Tin Whistle playing at ~6 sec



